Project Proposal

AI: Ethics, Policy & Society

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1. Project Title

Designing a responsible multi-agent safety architecture for Autonomous Vehicle Decision Systems.

2. Project Type

Technical Deliverable (15–20 page design brief with simulated scenario analysis)

3. Problem Statement

Autonomous vehicles make split-second decisions in situations where getting it wrong can cost lives. Yet most AV systems today operate as black boxes, there's no clear way to understand why a decision was made, who's responsible when something goes wrong, or how the system handles situations it's uncertain about.

This project explores how responsible AI principles can be embedded directly into an AV's decision-making architecture. Instead of treating safety and accountability as policy statements, we want to show they can be structural features of the system itself— built into how agents communicate, how uncertainty is handled, and when control gets handed back to a human.

The project works at the software and architecture level using simulated driving scenarios.

4. Domain

Urban autonomous vehicle navigation face high-uncertainty situations like busy intersections, unexpected pedestrians, poor weather, and sensor degradation. These are the moments where current AV systems are most vulnerable and where responsible design decisions matter most.

5. Framework Anchoring

- NIST AI RMF (2023)— Structures how risks are identified, measured, and mitigated across each agent using its Map-Measure-Manage functions

- EU AI Act— AVs fall under its high-risk AI category; informs the human oversight, logging, and auditability requirements in the governance design
- Jobin et al. (2019)— Provides cross-cutting AI ethics principles from global guidelines, used to evaluate whether the architecture actually delivers on its RAI claims
- Responsible Innovation (Owen et al.)— Informs the scenario analysis approach, anticipating failure modes at the design stage rather than after deployment

6. Methodological Approach

- Design a 4-agent architecture with clearly defined roles, inputs, and inter-agent communication
- Map RAI principles to specific components of the architecture and identify where gaps remain
- Apply consequence scanning across key failure scenarios (e.g., pedestrian not detected in rain)
- Use persona mapping to analyze the system from the perspective of a passenger, a pedestrian, a regulator, and a post-incident investigator
- Assess technical, socio-technical, and governance risks specific to multi-agent coordination

7. Deliverable Structure

1. Executive Summary
2. Problem and Context— the accountability gap in current AV systems and why architecture-level design matters
3. Multi-Agent Architecture Design— four agents (perception monitor, risk assessment, safety constraint enforcement, escalation/override) with a data flow diagram
4. RAI Principle Mapping— structured table linking each principle to a design mechanism and its limitations
5. Scenario Analysis— consequence scan and persona mapping across realistic failure cases
6. Risk Analysis— technical, human factors, and governance risks
7. Governance and Oversight Design— escalation logic, logging, and audit support
8. Reflection— honest assessment of what the design achieves and where it falls short

8. Anticipated Technical Contribution

This project produces a working reference for how RAI frameworks like the NIST AI RMF can be applied at the component level in a real-time autonomous system, not just at the organization or product level. It also surfaces an under-explored tension: distributing decisions across multiple agents can improve modularity but actually makes accountability harder to trace, which has real implications for how we govern autonomous systems going forward.

9. Preliminary Sources

1. NIST. (2023). *AI Risk Management Framework (AI RMF 1.0)*. National Institute of Standards and Technology.
2. Jobin, A., Ienca, M., & Vayena, E. (2019). The global landscape of AI ethics guidelines. *Nature Machine Intelligence*, 1(9), 389–399.
3. Owen, R., Macnaghten, P., & Stilgoe, J. (2012). Responsible research and innovation. *Science and Public Policy*, 39(6), 751–760.
4. European Parliament. (2024). *EU Artificial Intelligence Act*.
5. Grigorescu, S., et al. (2020). A survey of deep learning techniques for autonomous driving. *Journal of Field Robotics*, 37(3), 362–386.
6. Awad, E., et al. (2018). The Moral Machine experiment. *Nature*, 563, 59–64.
7. Rahwan, I. (2018). Society-in-the-loop. *Ethics and Information Technology*, 20(1), 5–14.
8. Varshney, K. R., & Alemzadeh, H. (2017). On the safety of machine learning. *Big Data*, 5(3), 246–255.

10. Workplan

Member	Responsibility
Avneet	Architecture design, data flow diagram, RAI principle mapping
Anish	Scenario analysis, consequence scan, persona mapping, risk and governance
Both	Reflection, integration, editing

Professor Kirkpatrick Comments

This is a strong proposal with a clear problem, a well-defined technical angle. The project is especially compelling in how it frames accountability, uncertainty, and escalation as design problems inside the autonomous vehicle system itself. The proposed four-agent structure gives the project a concrete backbone, and the use of consequence scanning and persona mapping adds useful socio-technical depth. One area to sharpen is the relationship between the conceptual architecture and the simulated scenarios. The proposal would benefit from stating more explicitly whether the main output is a design brief for a hypothetical architecture, a simulated evaluation of that architecture, or a comparative argument about why multi-agent design offers advantages over more conventional AV decision systems.

The methodology would be stronger with clearer definitions of what each agent is allowed to decide, what triggers escalation or override, and how success or failure will be assessed across scenarios. At present, the proposal has a strong structure, but the evaluation criteria remain somewhat general. It would help to specify a small set of realistic edge cases, identify what observable behaviors the system should exhibit in each one, and explain how traceability and accountability will actually be demonstrated rather than asserted. The source base is promising, though it would benefit from engagement with more directly relevant AV safety and human factors literature in addition to ethics and governance frameworks. Overall, this is a thoughtful and well-framed project with substantial potential, and it will be most persuasive if the architecture, scenario analysis, and evaluation logic are tied together with greater specificity.

Grade: 19